

Supporting Information: Lactic Acid Spectroscopy: Intra- and Intermolecular Interactions

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S1 ^1H -NMR on Lactic acid Sample

NMR Analysis: ^1H NMR experiments were obtained using a Bruker Fourier 300 NMR spectrometer operating at 300 MHz at 23°C.

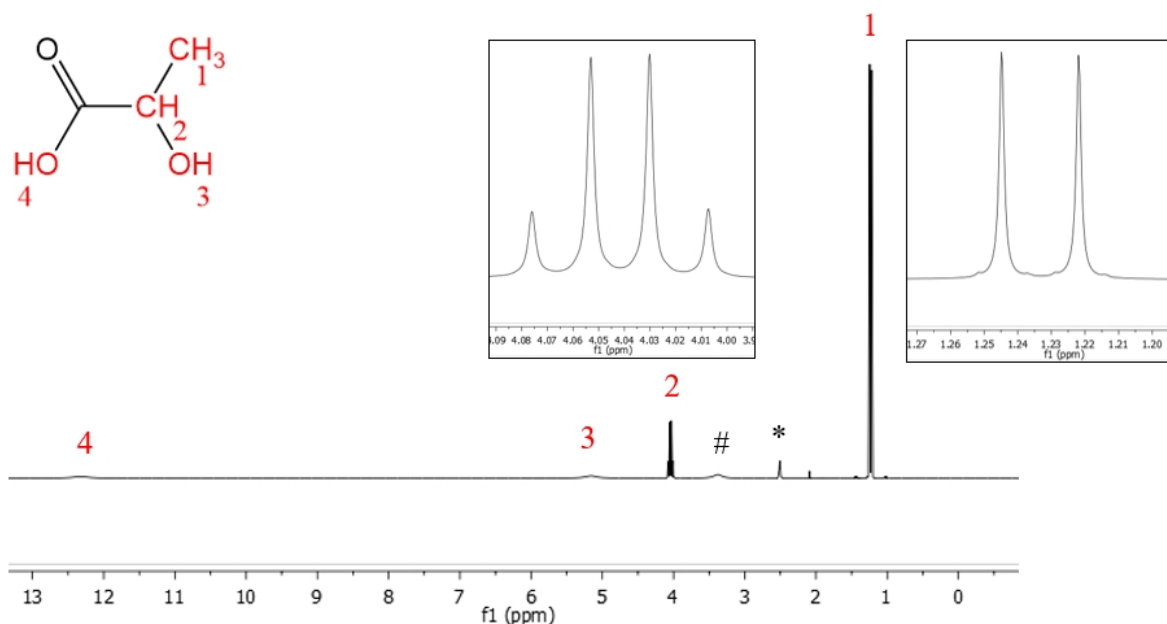


Figure S1: Sample ^1H -NMR of L-Lactic Acid in DMSO-d_6 with expanded spectral regions for peaks 1 and 2. Peaks numbered in red correspond with the H atoms in the lactic acid structure shown. The * indicates undeuterated DMSO and the # is a D_2O impurity in the DMSO-d_6 . The lack of significant unidentified peaks demonstrates the purity of the sample.

S2 Sample Cell for Gas-phase FTIR

An illustration of the cell used in the FTIR spectroscopy experiments is shown here in Figure S2. The CaF_2 window diameter is 2.54 cm and the thickness is 5 mm. The path length, as shown in the Figure, is 50 cm.

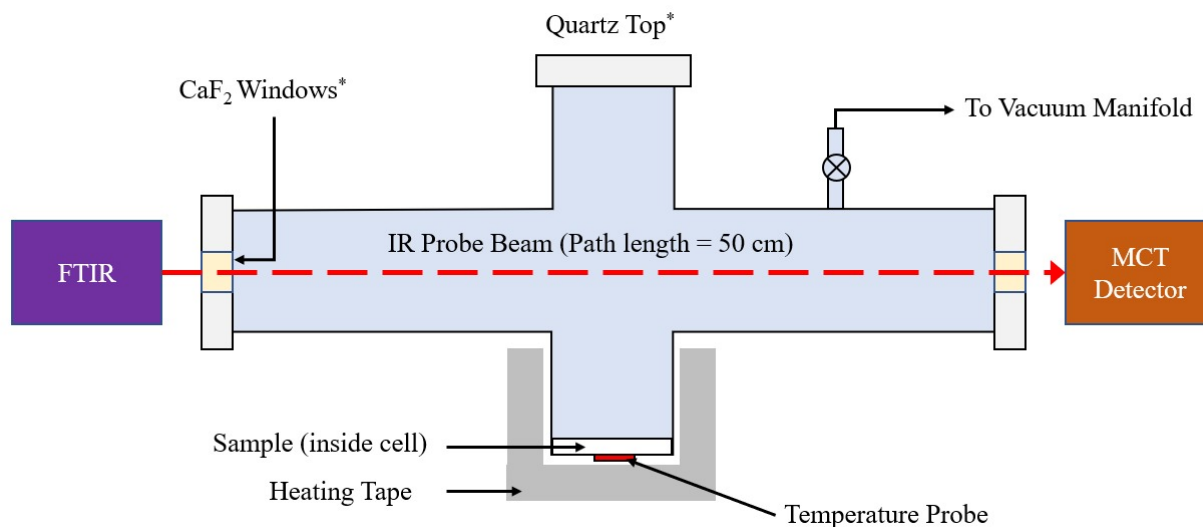
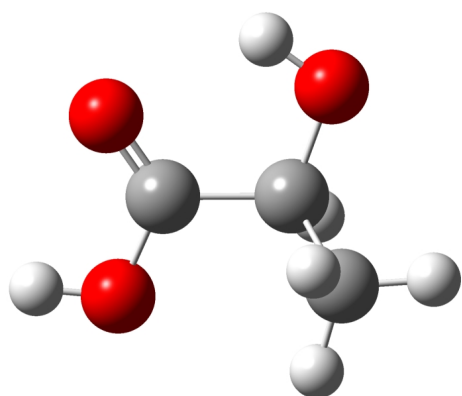
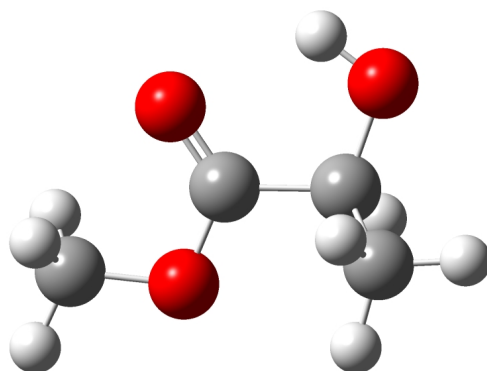


Figure S2: Sample cell used for gas-phase IR spectra collection. *The quartz top and CaF_2 windows are held in place using Viton O-rings and custom made PTFE clamps.

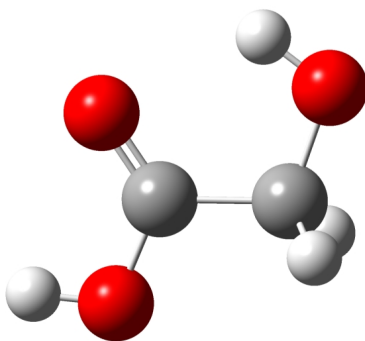
S3 Structure of Lactic Acid, Methyl Lactate and Glycolic Acid and Conformer Naming Convention



Lactic acid



Methyl Lactate



Glycolic acid

Figure S3: Structures of the lowest energy conformers of lactic acid, methyl lactate and glycolic acid.

The naming convention used for the conformers in this work follows what is already established in the literature.^[1,2] The shorthand terminology describes the three dihedral angles that define the conformation of lactic acid. the three angles include the alcohol H-O-C-C dihedral, the O-C-C=O dihedral and the acid H-O-C=O dihedral and are described by A=anti, S=syn, G=gauche, sk=skew, C=cis and T=trans. For example the SsC conformer is the "*Syn syn Cis*" conformer, the GskC conformer is the "*Gauche skew Cis*" conformer and the AaT conformer is the "*Anti anti Trans*" conformer. Note that the marks in G'sk'C indicate that the Gauche and skew dihedrals are slightly offset from the same dihedrals in the GskC conformer.

S4 Computational Method Benchmark on Glycolic Acid

In Table S1 the B3LYP/may-cc-pVTZ calculated vibrational transition frequencies, $\tilde{\nu}$, for glycolic acid are listed and compared to benchmark values obtained by Nejad *et al.*^[3] The benchmark values include both computational high level experimental numbers. In Table S2 the harmonic and anharmonic components of the the two OH stretch transitions in glycolic acid are compared with the work by Havey *et al.* who used overtone spectroscopy to uncover the harmonic and anharmonic components of the transition frequencies.^[4]

Table S1: Comparison between of our B3LYP/may-cc-pVTZ VPT2 method and the Nejad *et al.* coupled cluster calculations and experimental results for glycolic acid. Listed are the transition frequencies, $\tilde{\nu}_{\text{calc.}}$ in cm^{-1} and intensities, f , in parentheses in units of km/mol .

Vibrational mode	This work ^a	Nejad <i>et al.</i> ^[3]		
	$\tilde{\nu}_{\text{calc.}}, \text{cm}^{-1}$	^b $\tilde{\nu}_{\text{calc.}}, \text{cm}^{-1}$	^c $\tilde{\nu}_{\text{calc.}}, \text{cm}^{-1}$	^d $\tilde{\nu}_{\text{expt.}}, \text{cm}^{-1}$
ν_1 (OH acid)	3558.5	3573.7	3590.7	3586±2
ν_2 (OH alcohol)	3544.7	3570.6	3584.7	3578±2
ν_4 (C=O str.)	1770.6	1780.3	1788.9	1782±2
$\nu_9 + \nu_{12}$	1767.8	1779.5	1790.3	1797±2
ν_{11} (C-C str.)	838.1	847.7	853.2	856±2
ν_{12} (OCO bend)	638.7	637.9	643.6	648±2

^aVPT2 with B3LYP/may-cc-pVTZ. ^bHarmonic part at CCSD(T)/aug-cc-pVTZ level of theory with anharmonic correction at MP2/aug-cc-PVTZ level. ^cHarmonic part at CCSD(T)-F12a/cc-pVTZ-F12 level of theory with anharmonic correction at MP2/aug-cc-PVTZ level. ^dJet expansion Raman spectroscopy.

Table S2: Comparison of our B3LYP/may-cc-pVTZ VPT2 method with the Havey *et al.*^[4] experimental results for the two OH transitions in glycolic acid. Listed are the transition frequencies, $\tilde{\nu}$ in cm^{-1} , harmonic frequency, $\tilde{\omega}$, in cm^{-1} and anharmonic constant, $\tilde{\omega}x$, in cm^{-1} .

Vibrational mode	Source	$\tilde{\nu}, \text{cm}^{-1}$	$\tilde{\omega}, \text{cm}^{-1}$	$\tilde{\omega}x, \text{cm}^{-1}$
ν_1 (OH acid)	Havey <i>et al.</i> ^[4]	3582.4	3743.0	80.3
	This work (theory)	3558.5	3740.5	91.0
ν_2 (OH alcohol)	Havey <i>et al.</i> ^[4]	3583.3	3747.7	82.2
	This work (theory)	3544.7	3729.5	92.4

S5 List of Calculated Vibrational Transition Frequencies and Intensities

Here, the VPT2 calculated fundamental vibrational transitions are listed for the most abundant conformer of lactic acid, SsC, the 1:1 lactic acid:water complex SsC-H₂O (acid bound), and the lactic acid dimer SsC-SsC.

Table S3: The VPT2 calculated vibrational transition frequencies, $\tilde{\nu}$, and intensities, f , of the SsC conformer of lactic acid.

Mode	$\tilde{\nu}$ (cm ⁻¹)	f (km/mol)
ν_1	3558.97	107.63
ν_2	3543.29	33.53
ν_3	2983.77	16.69
ν_4	2996.10	14.82
ν_5	2934.35	5.62
ν_6	2863.12	6.57
ν_7	1763.03	221.51
ν_8	1457.85	0.79
ν_9	1450.71	3.86
ν_{10}	1392.32	6.37
ν_{11}	1367.32	2.59
ν_{12}	1331.90	24.16
ν_{13}	1268.55	44.19
ν_{14}	1237.00	6.10
ν_{15}	1152.20	9.87
ν_{16}	1109.77	252.41
ν_{17}	1076.36	33.47
ν_{18}	1024.03	56.40
ν_{19}	910.93	0.63
ν_{20}	797.25	14.59
ν_{21}	729.04	37.39
ν_{22}	665.50	8.70
ν_{23}	533.66	72.50
ν_{24}	493.12	10.14
ν_{25}	420.23	2.39
ν_{26}	375.94	22.03
ν_{27}	222.43	49.33
ν_{28}	305.54	19.43
ν_{29}	163.99	1.81
ν_{30}	70.23	1.65

Table S4: The VPT2 calculated vibrational transition frequencies, $\tilde{\nu}$, and intensities, f , of the SsC-H₂O acid complex.

Mode	$\tilde{\nu}$ (cm ⁻¹)	f (km/mol)
ν_1	3690.82	76.10
ν_2	3540.82	71.59
ν_3	3483.54	166.42
ν_4	3137.16	540.11
ν_5	2983.12	15.95
ν_6	2995.78	15.42
ν_7	2874.32	4.32
ν_8	2860.80	12.16
ν_9	1719.10	54.40
ν_{10}	1566.00	119.02
ν_{11}	1458.25	4.48
ν_{12}	1457.88	9.97
ν_{13}	1414.75	1.75
ν_{14}	1371.39	55.84
ν_{15}	1349.27	88.46
ν_{16}	1310.12	1.70
ν_{17}	1256.46	92.88
ν_{18}	1213.37	104.13
ν_{19}	1126.80	47.02
ν_{20}	1077.32	14.31
ν_{21}	1026.96	51.82
ν_{22}	916.04	1.31
ν_{23}	833.58	65.79
ν_{24}	810.45	21.33
ν_{25}	733.32	5.51
ν_{26}	626.27	2.37
ν_{27}	522.90	37.47
ν_{28}	502.04	73.88
ν_{29}	430.77	23.04
ν_{30}	158.37	3.30
ν_{31}	328.92	3.12
ν_{32}	473.18	134.24
ν_{33}	240.85	19.28
ν_{34}	206.68	91.72
ν_{35}	271.75	69.54
ν_{36}	182.80	11.06
ν_{37}	132.94	11.90
ν_{38}	77.25	0.22
ν_{39}	50.90	3.30

Table S5: The VPT2 calculated vibrational transition frequencies, $\tilde{\nu}$, and intensities, f , of the SsC-SsC dimer.

Mode	$\tilde{\nu}$ (cm ⁻¹)	f (km/mol)	Mode	$\tilde{\nu}$ (cm ⁻¹)	f (km/mol)
ν_1	3561.45	0.38	ν_{34}	1076.63	29.18
ν_2	3561.27	139.04	ν_{35}	1025.07	78.26
ν_3	2870.53	685.48	ν_{36}	1026.61	5.09
ν_4	2987.82	2.64	ν_{37}	966.09	152.38
ν_5	2987.96	28.54	ν_{38}	931.16	0.45
ν_6	2995.56	0.55	ν_{39}	913.03	2.96
ν_7	2995.74	50.35	ν_{40}	915.61	0.07
ν_8	2730.56	0.11	ν_{41}	828.92	26.90
ν_9	2946.27	16.31	ν_{42}	819.22	0.04
ν_{10}	2947.33	0.76	ν_{43}	754.61	4.97
ν_{11}	2854.31	32.28	ν_{44}	743.78	2.02
ν_{12}	2853.61	39.36	ν_{45}	634.73	30.56
ν_{13}	1701.97	476.55	ν_{46}	623.75	0.33
ν_{14}	1633.75	0.10	ν_{47}	525.08	10.09
ν_{15}	1469.17	0.01	ν_{48}	507.45	0.03
ν_{16}	1465.41	24.28	ν_{49}	410.23	12.53
ν_{17}	1463.50	0.51	ν_{50}	441.98	75.34
ν_{18}	1460.60	24.44	ν_{51}	275.41	46.63
ν_{19}	1439.37	2.80	ν_{52}	378.43	76.73
ν_{20}	1431.75	5.44	ν_{53}	220.76	24.45
ν_{21}	1360.69	9.38	ν_{54}	273.35	4.42
ν_{22}	1375.03	7.30	ν_{55}	453.24	8.63
ν_{23}	1371.03	2.99	ν_{56}	234.26	17.78
ν_{24}	1355.92	128.84	ν_{57}	183.27	0.27
ν_{25}	1313.30	24.66	ν_{58}	194.16	0.06
ν_{26}	1315.36	2.75	ν_{59}	130.67	0.002
ν_{27}	1276.33	216.97	ν_{60}	118.55	0.67
ν_{28}	1275.51	0.08	ν_{61}	110.70	11.39
ν_{29}	1225.34	0.82	ν_{62}	132.06	1.56
ν_{30}	1224.47	74.80	ν_{63}	128.89	4.01
ν_{31}	1132.75	0.24	ν_{64}	30.83	0.02
ν_{32}	1130.93	90.89	ν_{65}	15.06	0.89
ν_{33}	1077.70	5.17	ν_{66}	50.99	4.15

S6 Water Ice in the Experimental Spectra

Water ice absorption peaks are routinely observed in our spectra due to condensation on our external MCT detector. The MCT detector is cooled by liquid nitrogen, which causes water to freeze onto the window to the MCT detector. The water ice spectrum from Warren *et al.*^[5] is compared to our lactic acid spectrum to better visualize which signals belong to water ice, see Figure S4. Note that the Warren *et al.*^[5] spectrum is nominally at $T=266$ K while the water ice in our spectrum is of an uncertain temperature but likely closer to the boiling point of N_2 at $T\sim 80$ K. The water ice spectrum has been scaled to fit the peak at ~ 3300 cm^{-1} . The baseline correction of our experimental lactic acid gas phase spectrum seems to introduce a slight asymmetry to the dominant water ice peak. While the peak at ~ 3300 cm^{-1} is by far the most significant, the weaker signal at 1630 cm^{-1} may contribute to the broad feature on the left shoulder of the intense C=O stretching transition peak. Also note CO_2 is observable in our spectrum at ~ 2350 cm^{-1} . The intensity fluctuates throughout measurements indicating a slow leak into the cell or changed outside of the cell.

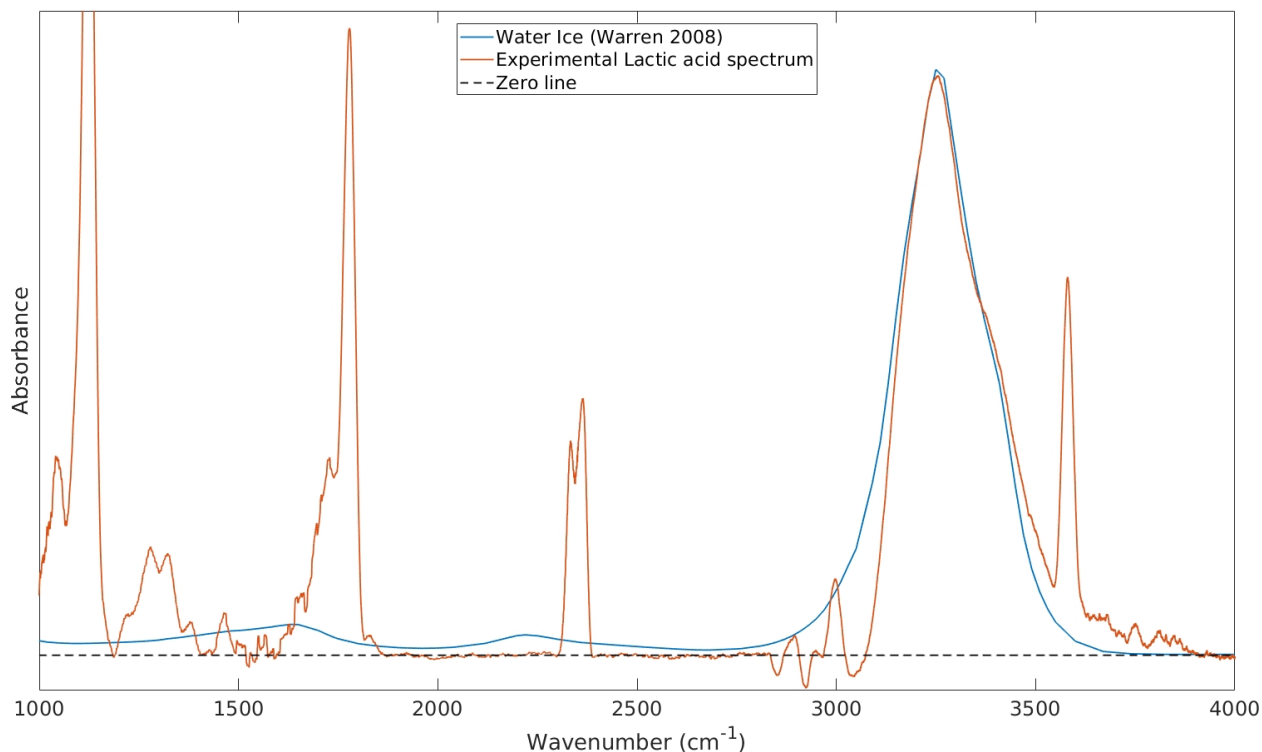


Figure S4: Water ice spectrum from Warren *et al.*^[5] compared to our gas phase lactic acid spectrum.

S7 Gaussian Function Fitting of IR Spectra

The lactic acid infrared spectra with fitted Gaussian functions are shown here in Figure S5, S6, S7, S8 and S9. An overview of the fitted function parameters is given in Table S6.

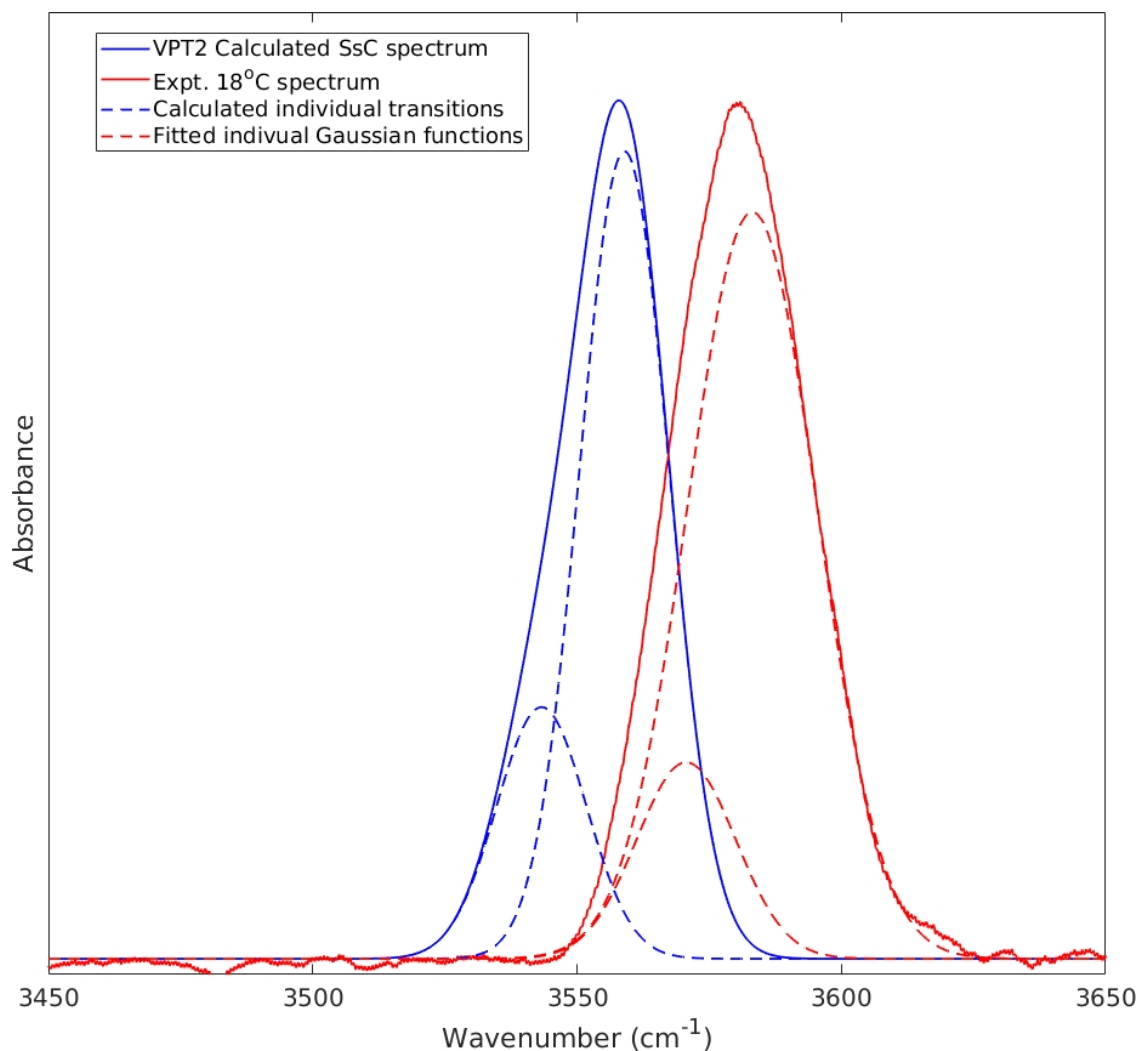


Figure S5: Experimental lactic acid infrared spectrum (red) compared to the calculated spectrum (blue) in the OH stretch region. Calculated individual transitions (dashed blue) and Gaussian functions fitted to the experimental signal (dashed red) are also shown.

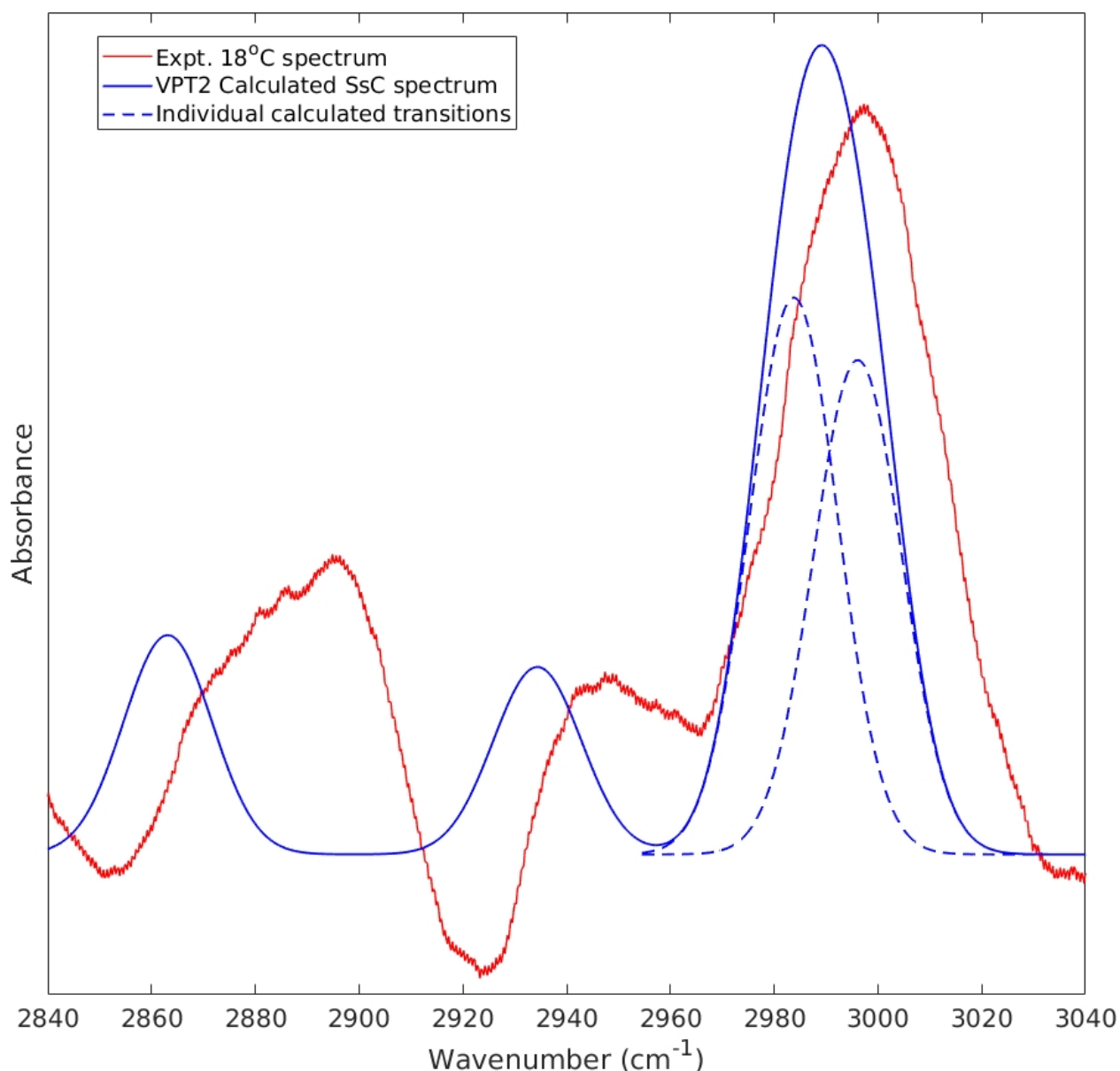


Figure S6: Lactic acid infrared spectrum (red) compared to the calculated spectrum (blue) for the CH stretch region. The dashed lines show the calculated individual transitions that make up the total spectrum for the band centered at 2997 cm^{-1} .

No fitting was done to peaks in the CH stretch region due to the skewed baseline. There are three observable bands and four calculated transitions for the SsC conformer in this range. As can be seen in Figure S6, two of the calculated transitions can plausibly merge to form a single band. Only one band is observed in the as is observed. The experimentally observed band maxima are located at 2997 cm^{-1} , 2947 cm^{-1} and 2895 cm^{-1} , while the calculated transition energies are 2996 cm^{-1} ,

2984 cm^{-1} , 2934 cm^{-1} and 2863 cm^{-1} .

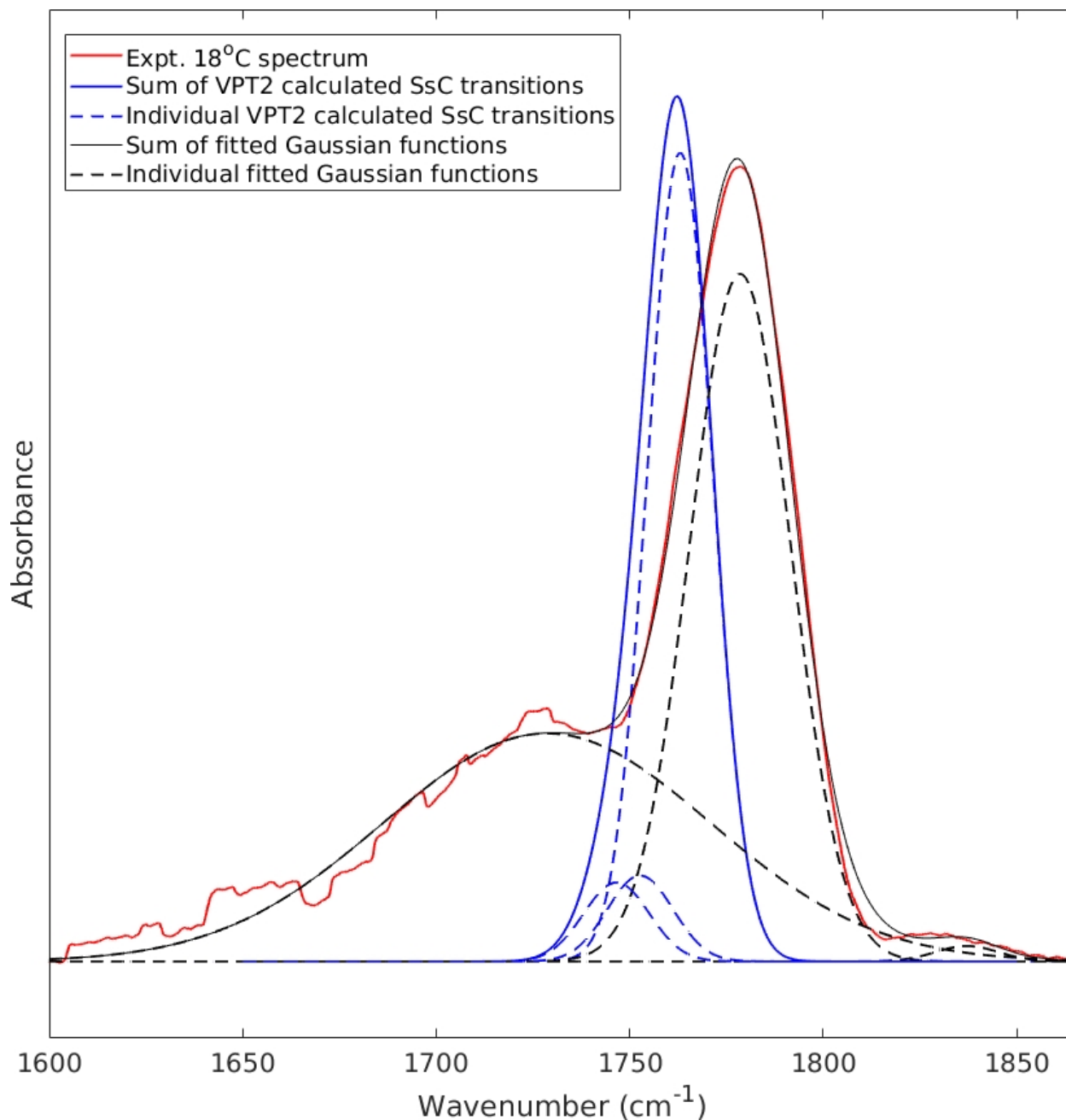


Figure S7: Lactic acid infrared spectrum (red) compared to the calculated spectrum (blue) and fitted Gaussian functions to the experimental signal (black) for the CO stretch region. The dashed lines show the calculated individual transitions which make up the solid line spectra of the same color.

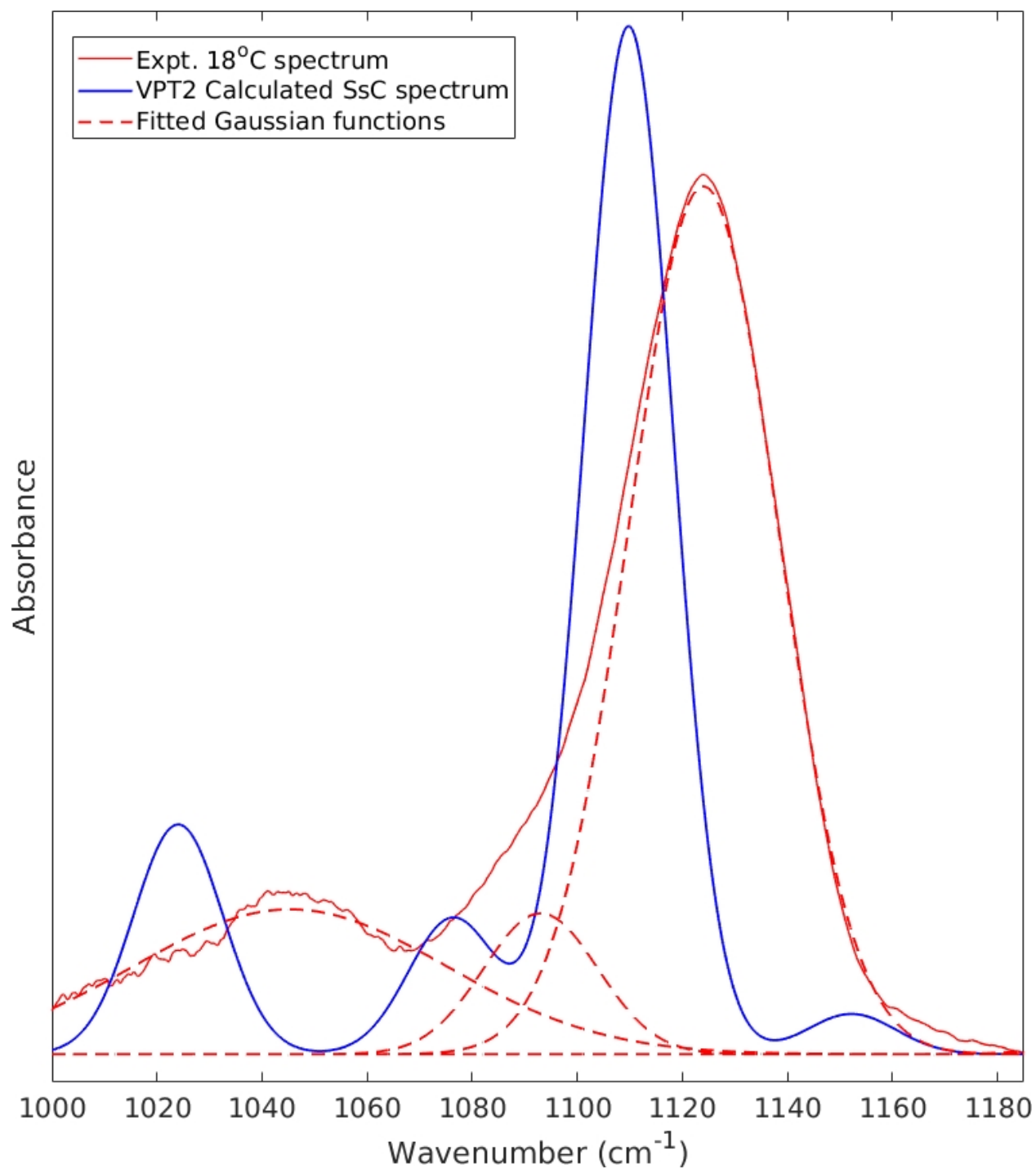


Figure S8: Lactic acid infrared spectrum (red) compared to the calculated spectrum (blue) and Gaussian functions fitted to the experimental signals (red dashed) in the fingerprint/skeletal mode region (1000-1180 cm⁻¹).

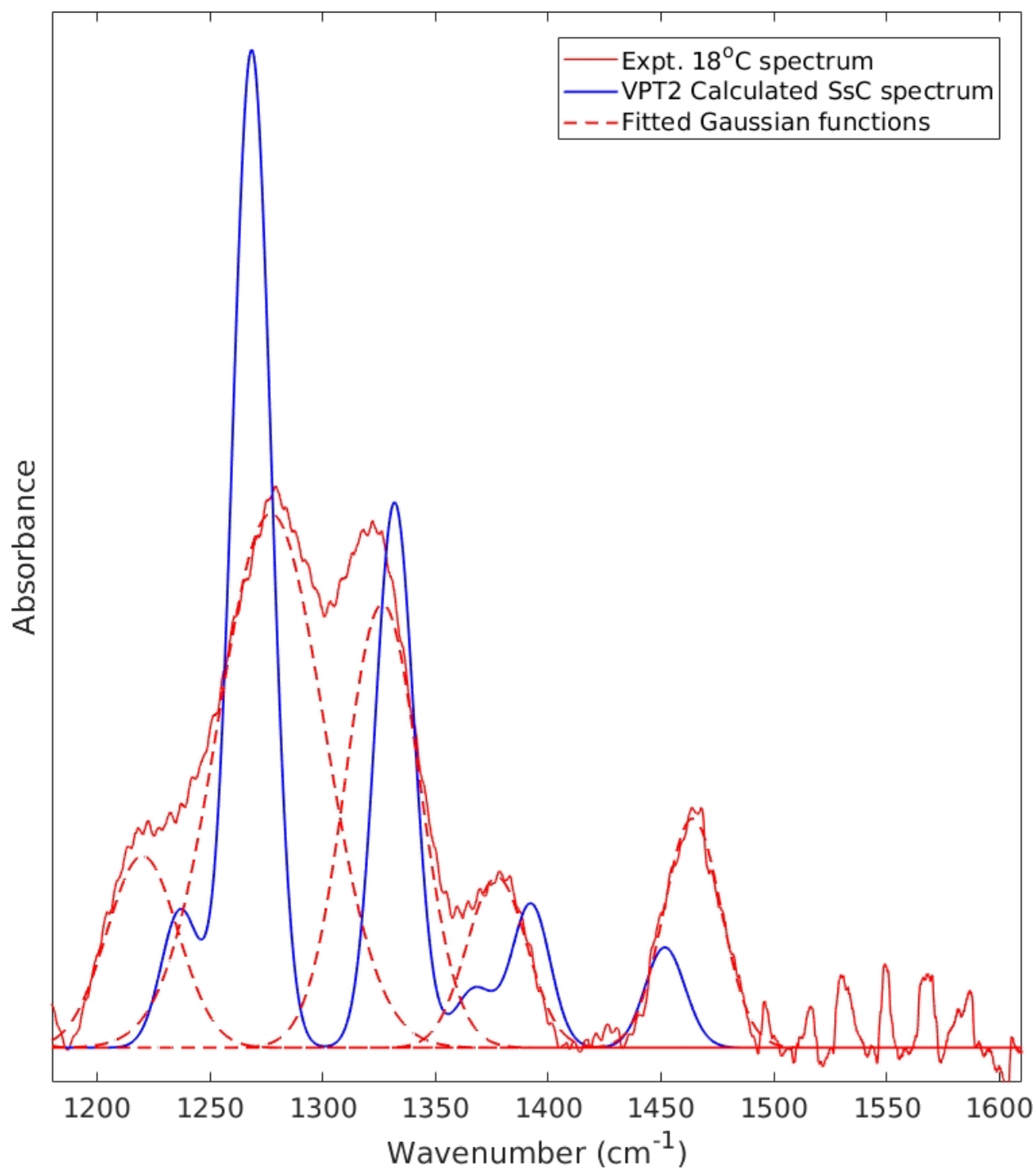


Figure S9: Lactic acid infrared spectrum (red) compared to the calculated spectrum (blue) and Gaussian functions fitted to the experimental signals (red dashed) in the fingerprint/skeletal mode region (1180-1600 cm^{-1}).

It can be challenging to assign individual transitions in the fingerprint region since there are many different peaks observed. However, in general there seems to be a calculated transition that approx-

imately corresponds to an observed peak in all cases, see Figures S8, S9 and Table S3. Especially in Figure S9, there are five fitted peaks which seem to correspond nicely to six calculated transitions. The calculated transition at 1367 cm^{-1} is the only one not observed and is likely weak enough to go unnoticed in the experimental spectrum under the feature from $1200\text{--}1400\text{ cm}^{-1}$. The same can be said for Figure S8, though with less certainty due to the overlap of the 1077 and 1130 cm^{-1} . Additionally, the since one signal is buried under the 1124 cm^{-1} feature and the 1045 cm^{-1} band is significantly broader than other observed lactic acid bands which may suggest it is not originating from a single vibrational transition in the SsC conformer of lactic acid. Experimentally fitted bands in Table S6 can be directly compared to the calculated transitions in Table S3 to make these assignments.

Table S6: Parameters of the fitted Gaussian functions the lactic acid spectrum at 18°C , assigned to the SsC conformer unless indicated otherwise. The parameters are the peak maximum ($\tilde{\nu}$), the full width at half maximum (FWHM) and the integrated absorbance ($\int A(\tilde{\nu})$). The listed uncertainties are only from the fitting procedure and does not take any other sources of uncertainty into account. Skeletal modes are numbered in terms of descending transition wavenumber.

Vib. mode assignment	$\tilde{\nu}\text{ (cm}^{-1}\text{)}$	FWHM (cm^{-1})	$\int A(\tilde{\nu})\text{ (cm}^{-1}\text{)}$
OH str. (acid)	3583 ± 0.9	29 ± 0.7	$(5.09\pm0.44)\times10^{-2}$
OH str. (alcohol)	3571 ± 0.8	22 ± 1.4	$(1.01\pm0.44)\times10^{-2}$
C=O str. high energy conformer(s) ^a	1837 ± 1.2	19 ± 3.0	$(0.14\pm0.02)\times10^{-2}$
C=O str.	1779 ± 0.03	31 ± 0.1	$(10.10\pm0.07)\times10^{-2}$
Skeletal mode 1	1464 ± 0.2	30 ± 0.4	$(0.84\pm0.01)\times10^{-2}$
Skeletal mode 2	1377 ± 0.3	31 ± 0.7	$(0.66\pm0.01)\times10^{-2}$
Skeletal mode 3	1327 ± 0.3	39 ± 0.7	$(2.12\pm0.06)\times10^{-2}$
Skeletal mode 4	1277 ± 0.3	56 ± 1.3	$(3.66\pm0.08)\times10^{-2}$
Skeletal mode 5	1220 ± 0.5	37 ± 0.9	$(0.88\pm0.03)\times10^{-2}$
Skeletal mode 6	1124 ± 0.1	34 ± 0.1	$(19.92\pm0.06)\times10^{-2}$
Skeletal mode 7	1093 ± 0.2	26 ± 0.5	$(2.47\pm0.09)\times10^{-2}$
Skeletal mode 8	1045 ± 0.3	70 ± 0.7	$(6.95\pm0.06)\times10^{-2}$

^aPotential assignment to GskC, G'sk'C or the AaT conformer, with more than one conformer possibly contributing.

Uncertainties are significantly higher than indicated by those on the fit here.

S8 CaF₂ Windows After Spectra Collection

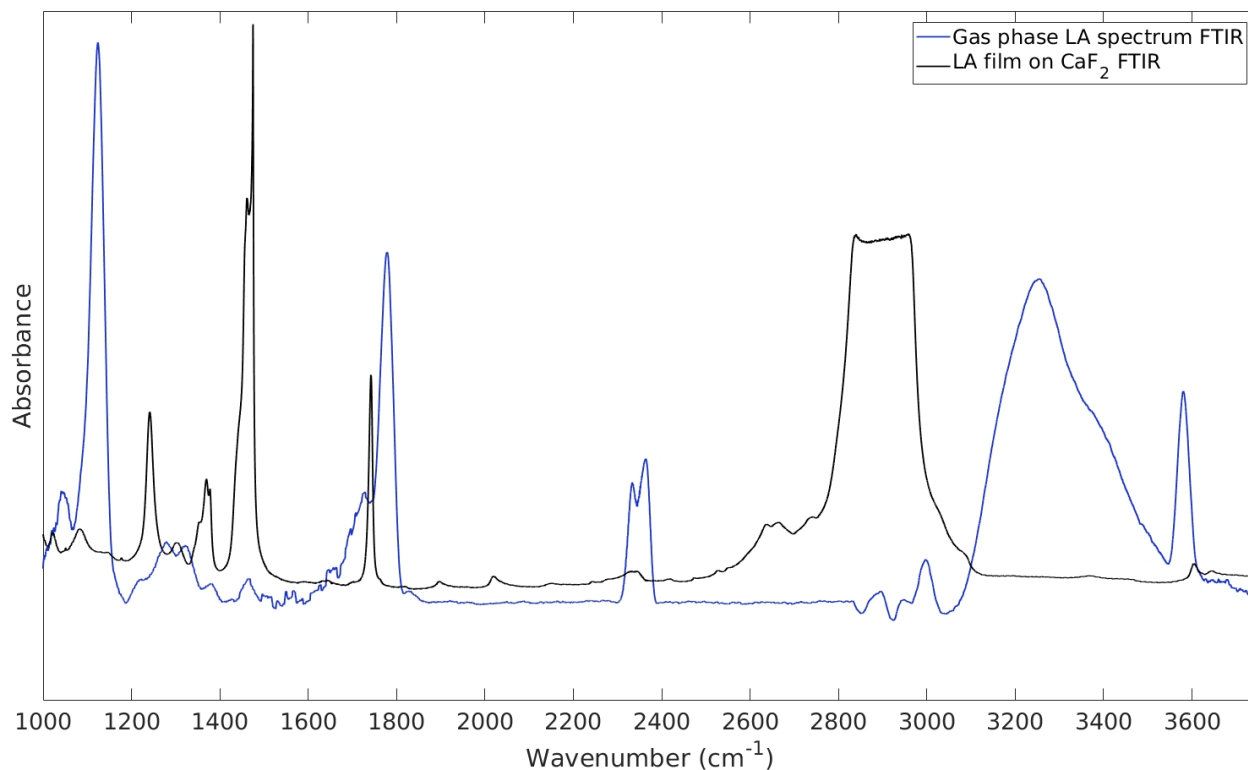


Figure S10: Comparison between the FTIR spectrum of lactic acid in the gas phase (blue) and a spectrum taken of the CaF₂ windows after gas phase spectrum collection (black). The absorbance in the blue spectrum has been multiplied by a factor 500 to better compare the two spectra. The peak at $\sim 2900\text{ cm}^{-1}$ in the black spectrum is saturated which is why it has a flat top. LA is shorthand for lactic acid.

The CaF₂ window spectrum was obtained by first recording a background spectrum of the windows used in the gas phase cell prior to assembling the cell for recording a gas phase lactic acid spectrum. Then after having run the lactic acid gas phase experiment, the cell was disassembled and the final spectrum of the windows was recorded. Figure S10 shows that materials deposited on the windows do not add significantly to the lactic acid gas phase spectrum.

S9 Supplemental NCI/Bonder Figure

Figure S11 plots the Gibbs energy versus the calculated $G(s_{0.5})$ value of the internal hydrogen bond in the different lactic acid conformers. Note that $G(s_{0.5})$ values are set to zero for the conformers where the alcohol functional group is not a hydrogen bond donor. The AaT conformer does have an internal hydrogen bond with the acid as the donor and alcohol as the acceptor, with a calculated $G(s_{0.5})$ value of 0.017, which if included in the plot would place it as firmly outside of the implied correlation as a value of zero does. Figure S11 is an attempt to show that two things dominate the relative energy of the lactic acid conformers: 1) Orientation of the carboxylic acid group, with the *cis* conformation being favored and 2) the strength of the internal hydrogen bond. Caution is, however, warranted in such an interpretation since Gibbs energies depend on more factors than the two mentioned.

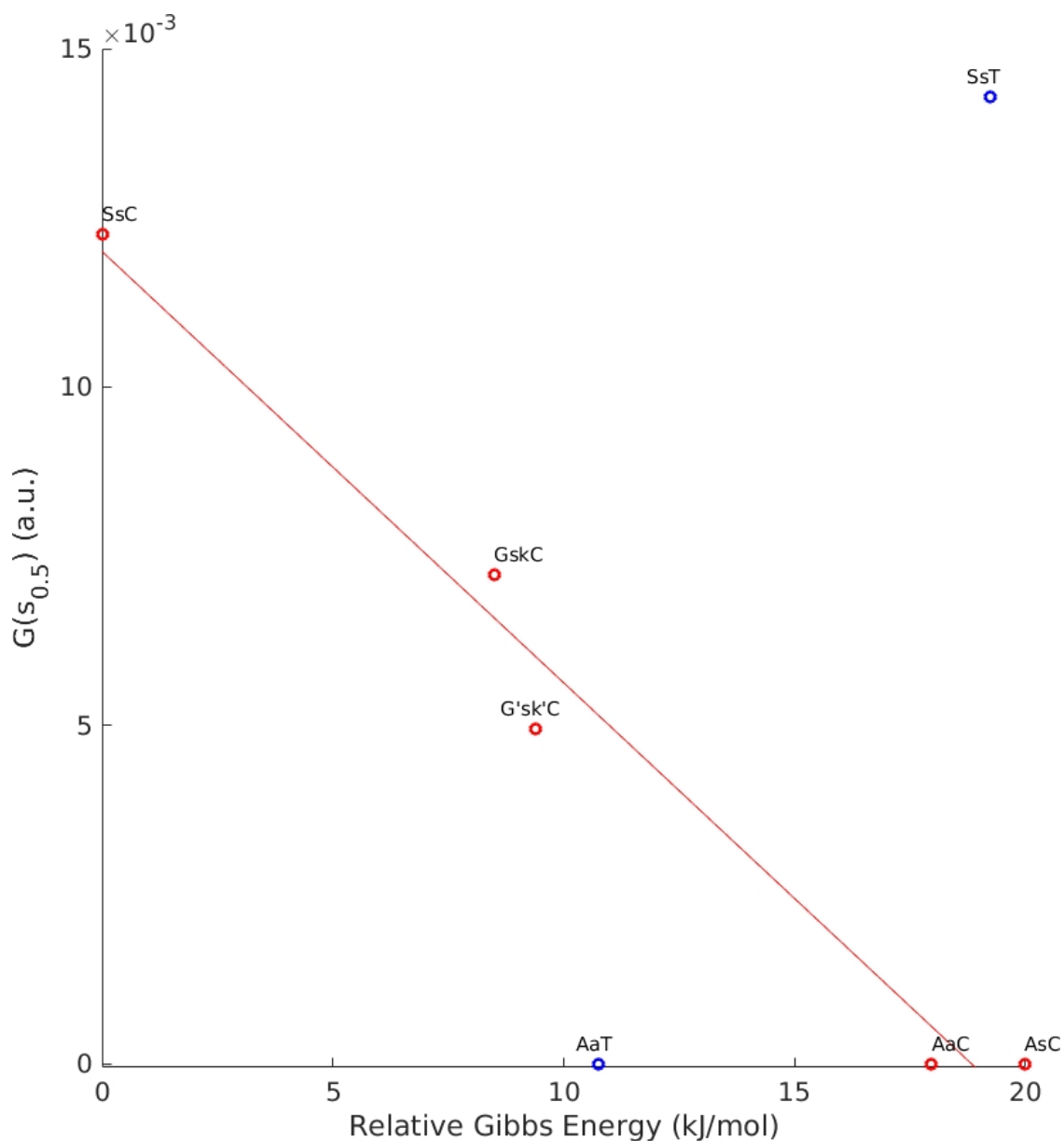


Figure S11: The $G(s_{0.5})$ value vs. the relative Gibbs energy (ΔG) for the six lactic acid conformers. The five *cis* conformers in red appear to display a correlation between the two values, indicating that internal hydrogen bond strength affect the relative stability of the conformers significantly. The two *trans* conformers in blue do not fit in this correlation.

S10 Computational Methods Details

As a first step geometry optimization followed by normal mode calculations were done with the described DFT method, to identify the ground state geometries of the investigated molecule and to obtain the harmonic frequencies of each minimum energy structure. The DFT calculations in this work with the B3LYP and B3LYP-D3 functionals included the keywords "opt=vtight" and "Integral(Grid=SuperFine)", while the calculations with the ω B97X-D functionals did not. The "opt=vtight" keyword is used to impose more stringent convergence criteria during the geometry optimizations, while the "Integral(Grid=SuperFine)" keyword increases the amount of points in the integration grid used for DFT calculations, improving accuracy slightly. The calculated frequencies were checked to see if they were all positive, to ensure that a minimum on the potential energy surface had been found, and any negative frequencies in the output would indicate that the geometry optimizations had not identified a local minimum on the potential energy surface. All calculated frequencies were found to be positive. The thermochemical properties of investigated molecules are automatically obtained in output files created by the Gaussian program. The thermodynamic properties were calculated with default settings in Gaussian and without anharmonic corrections.^[6]

The choice of the may-cc-pVTZ basis set rested on preliminary testing of the cc-pVTZ basis set with the prefixes "aug-", "jul-", "jun-" and "may-" on the lactic acid monomers using the B3LYP functional. Relative Gibbs energies between conformers were calculated using the aug-cc-pVTZ and may-cc-pVTZ basis sets and the calculated relative Gibbs energies were identical within 0.2 kJ/mol for the five most stable conformers (SsC, GskC, G'sk'C, AaT and AaC). This is an insignificant variation compared to the computed relative Gibbs energies which are on the order of 8 and 9 kJ/mol for SsC compared to GskC and G'sk'C. VPT2 calculations were carried out for the SsC conformer with all the mentioned basis sets to test for the basis set influence on the vibrational transition energies and intensities. Relative peak positions and intensities for all fundamental transitions are generally very close to each other in all four basis sets. In terms of absolute transition energies, the $>500\text{ cm}^{-1}$ fundamental transition frequencies do in most cases fall within 5 cm^{-1}

and a few of them within 10 cm^{-1} , across the four basis sets. The biggest exception is the calculated anharmonic frequency for the CH_3 symmetric stretch, which for the "aug-" and "jul-" basis sets lies at 2863 cm^{-1} and 2864 cm^{-1} , respectively, and for the "jun-" and "may-" basis sets lies at 2936 cm^{-1} and 2934 cm^{-1} , respectively. Note that the anharmonic frequency for the transitions is within 1 cm^{-1} of each other for all four basis sets, so this is clearly an effect of including anharmonicity with VPT2. The reasonable agreement with relative peak placements in the CH region of the experimental spectra to the B3LYP/may-cc-pVTZ serve as a validation for using this basis set over the larger "aug-" and "jul-" prefixes for the cc-pVTZ basis set. Furthermore, using the may-cc-pVTZ basis set took 2.5 times fewer CPU hours for the VPT2 calculations compared to the aug-cc-pVTZ for the lactic acid monomer. This made the may-cc-pVTZ basis set our preferred basis set for further calculations on the larger systems such as the 1:1 lactic acid water complexes, and the lactic acid dimer.

The $\omega\text{B97X-D}$ functional was also tested using the aug-cc-pVTZ basis set. Unfortunately, the VPT2 calculations with $\omega\text{B97X-D}$ gave unrealistic results for some of the vibrational modes, such as large negative transitions frequencies and unusually large transition intensities. The harmonic frequencies and intensities were entirely reasonable. Thus, for the purpose of including anharmonicity in the computed vibrational modes in this work, the $\omega\text{B97X-D}$ functional was not useful. For the purpose of calculating relative energies between conformers and complexation energies we used $\omega\text{B97X-D/aug-cc-pVTZ}$, B3LYP-D3/may-cc-pVTZ, B3LYP/may-cc-pVTZ and finally CCSD(T)/cc-pVTZ//B3LYP/may-cc-pVTZ, the latter method denotes using B3LYP/may-cc-pVTZ to obtain optimized structures and thermodynamic parameters and then calculating the electronic energy at the CCSD(T)/cc-pVTZ level of theory. The comparison of relative energies between conformers are summarized in Table S7. The calculated relative Gibbs energies appear to be in good agreement across methods, with the exception of some of the higher energy water complex conformers, which are not of significance to this work. The comparison of complexation energies for the SsC conformer are shown in Table S8.

Table S7: Relative Gibbs energy (ΔG) between the lactic acid conformers in kJ/mol calculated using four different methods calculated at 18°C. For alcohol we use the shorthand "alc." and for the basis sets we use the following shorthand: cc-pVTZ = VTZ, may-cc-pVTZ = mVTZ and aug-cc-pVTZ = aVTZ. This table includes lactic acid monomers, dimers and water complexes separated by horizontal lines, and energies are to be interpreted as scaled relative to the lowest energy conformer for each disjointed set.

Conformer	CCSD(T)/VTZ// B3LYP/mVTZ	B3LYP/mVTZ	B3LYP-D3/mVTZ	ω B97X-D/aVTZ
SsC	0.0	0.0	0.0	0.0
GskC	8.5	8.1	8.2	8.7
G'sk'C	9.4	9.0	8.9	9.8
AaT	10.8	10.1	10.3	9.3
AaC	18.0	17.0	17.5	17.4
SsT	19.2	19.6	19.5	18.6
AsC	20.0	18.1	18.7	18.8
(SsC) ₂	0.0	0.0	0.0	0.0
SsC-GskC	6.4	5.9	6.5	6.5
SsC-G'sk'C	6.5	6.3	6.6	6.2
(GskC) ₂	12.6	11.7	12.4	12.6
GskC-G'sk'C	12.7	12.0	12.6	11.2
SsC-AaC	16.9	16.4	17.3	16.0
SsC-H ₂ O acid	0.0	0.0	0.0	0.0
GskC-H ₂ O acid	8.1	7.6	7.8	9.2
G'sk'C-H ₂ O acid	8.2	8.0	8.0	8.6
AsC-H ₂ O acid	18.9	18.8	19.5	18.7
SsT-H ₂ O acid	19.3	19.0	20.4	30.7
AaC-H ₂ O acid	19.5	18.7	19.2	18.6
GskC-H ₂ O alc.	22.6	24.2	22.9	23.7
SsC-H ₂ O alc	29.7	12.8	11.0	11.5
AaT-H ₂ O acid	30.4	29.7	30.6	30.9
SsT-H ₂ O alc.	33.0	33.6	31.6	30.7

Table S8: Complexation energy for formation of the SsC-SsC dimer and the SsC-H₂O complex in kJ/mol calculated using four different methods calculated at 18°C. A negative Gibbs energy signifies an attractive complexation energy, in other words a greater than unity equilibrium constant. For the basis sets we use the following shorthand: cc-pVTZ = VTZ, may-cc-pVTZ = mVTZ and aug-cc-pVTZ = aVTZ.

Conformer	CCSD(T)/VTZ// B3LYP/mVTZ	B3LYP/mVTZ	B3LYP-D3/mVTZ	ω B97X-D/aVTZ
SsC-SsC	-21.6	-12.5	-24.5	-20.6
SsC-H ₂ O	-5.0	5.3	-1.5	1.3

Wavefunction Analysis Supplemental Information

The resolution settings in the Bonder program were tested at resolutions of 0.20, 0.10, 0.05, 0.04 and 0.02. The resolutions of 0.20 and 0.10 are too coarse as $G(s_{0.5})$ values changed significantly with higher resolutions. From 0.04 to 0.02 resolution the resulting $G(s_{0.5})$ did not change in any significant manner (less than 1%) and thus we decided to use 0.04 resolution for these calculations as a good compromise between CPU time and accuracy.

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